The Journal of Open Source Software

MEXsrfdcrPy: Spatial Random Forests for Daily Climate Records Reconstruction in Mexico

Hugo Antonio-Fernández [1,2], **Humberto Vaquera-Huerta** [1], **Moisés Michel Rosengaus-Moshinsky** [3], **Paulino Pérez-Rodríguez** [1], **and José Crossa** [4]

1 Colegio de Postgraduados, México **2** Universidad Mexiquense del Bicentenario, México **3** Independent Consultant **4** CIMMYT, México

DOI: 10.xxxxxx/draft

Software

- Review
- Repository
- Archive

Editor:

Submitted: 11 December 2025
Published: unpublished

License
Authors of papers retain copyright and release the work under a Creative Commons Attribution 4.0 International License (CC BY 4.0).

DRAFT

Summary

Daily climate station records underpin hydrological design, agricultural planning and climate-risk assessment, yet long series are almost always affected by gaps, relocations and inconsistent maintenance (Menne et al., 2012). In México, the national network maintained by the Servicio Meteorológico Nacional (SMN) comprises thousands of stations with heterogeneous length and completeness, complicating nationwide analyses and the calibration of gridded products (Comisión Nacional del Agua (CONAGUA), 2012).

MEXsrfdcrPy is a Python package that addresses this problem by training **one global random forest model** on all available daily observations using only geographic coordinates and calendar information—latitude, longitude, elevation and time—and then using this model to reconstruct missing values, quantify spatial interpolation skill station by station, and generate climate grids at user-defined resolutions. The same workflow applies to daily precipitation, minimum and maximum temperature, and evaporation.

The package builds on NumPy, pandas and scikit-learn (Harris et al., 2020; McKinney, 2010; Pedregosa et al., 2011) and is designed for interactive notebooks, scripted workflows and large-scale jobs on platforms such as Kaggle. MEXsrfdcrPy forms the spatial interpolation layer, with calendar-based temporal features, of a broader open-source ecosystem for Mexican climate data, complementing tools for station-data access and local spatial–temporal imputation (Antonio-Fernández et al., 2025c, 2025b).

Statement of need

Spatio-temporal interpolation of station data is commonly performed with kriging, inverse distance weighting or thin-plate splines (Hijmans et al., 2005; Hofstra et al., 2008). These methods can yield high-quality maps but often require variable-specific configuration and explicit covariance modeling. In parallel, machine-learning approaches—particularly random forests—have become popular for spatial prediction because they capture nonlinear responses and interactions without strong distributional assumptions (Breiman, 2001; Hengl et al., 2018). Many such applications, however, rely on rich sets of auxiliary predictors that are not always available at station locations or for long historical periods.

MEXsrfdcrPy targets a complementary use case: **reconstructing daily station series and generating gridded fields using only coordinates and time**. The package assumes that a substantial fraction of the climatological signal can be learned from latitude, longitude, elevation and day of year when training data aggregate decades of observations from a dense national network. This design makes the method easy to deploy across variables and periods, avoids dependence on external reanalyses or remote-sensing products, and yields reusable

global models that can be archived with DOIs and shared between projects. The approach is suited to national and regional studies that require transparent reconstruction of long daily series for crop modeling, drought analysis or climate-change attribution (Ruane et al., 2015; Xu et al., 2024).

Model and implementation

At its core, MEXsrfdcrPy fits a spatial random forest to a long-format daily table with station identifier, date, latitude, longitude, elevation and a single numeric target. Time is represented through year, month and day of year, with optional sinusoidal terms for the annual cycle. A single `RandomForestRegressor` (Breiman, 2001; Pedregosa et al., 2011) is trained using all stations that exceed a user-defined threshold of valid observations within a specified training window, and the fitted model plus metadata—column names, feature set, training period and station coverage—are persisted as joblib and JSON artifacts for reproducibility and reuse.

The package is organized around two high-level modules. The **`loso`** module implements leave-one-station-out (LOSO) experiments: each station is excluded from training and then predicted across the evaluation period. Results are summarized in tidy tables with three metrics—mean absolute error (MAE), root mean squared error (RMSE) and the coefficient of determination (R^2)—computed on daily values and on user-defined temporal aggregations, following best practices for spatial cross-validation (Roberts et al., 2017). The **`grid`** module reuses a single global model to predict any station grid or regular mesh defined by latitude, longitude and elevation, over any compatible date interval. Internally, the package uses NumPy arrays for fitting and prediction and delegates parallelization to scikit-learn's multi-core implementation, keeping dependencies minimal and installation straightforward.

Examples

MEXsrfdcrPy is illustrated using a 1991–2020 SMN daily dataset for México (Antonio-Fernández et al., 2025a). A typical analysis first runs LOSO experiments over a region to quantify station-by-station MAE, RMSE and R^2, and then trains a global model on the full national network to generate continuous fields on a fine grid for the same period. (**?**) shows a representative station-level comparison produced directly with the package plotting utilities. Daily rainfall at an SMN station is contrasted with three alternative sources of information: an external gridded product from NASA POWER (variable PRECTOTCORR) (National Aeronautics and Space Administration, 2020), a local spatial–temporal random forest model obtained with MissClimatePy (Antonio-Fernández et al., 2025b), and a grid-based global model derived from MEXsrfdcrPy applied to a high-resolution mesh. The legend reports MAE, RMSE and R^2 for each series, showing that, at this station, the global spatial random forest can match or outperform both the external product and the local model. The figure thus illustrates the dual role of MEXsrfdcrPy as a reconstruction engine and as a benchmarking tool against existing datasets.

DRAFT

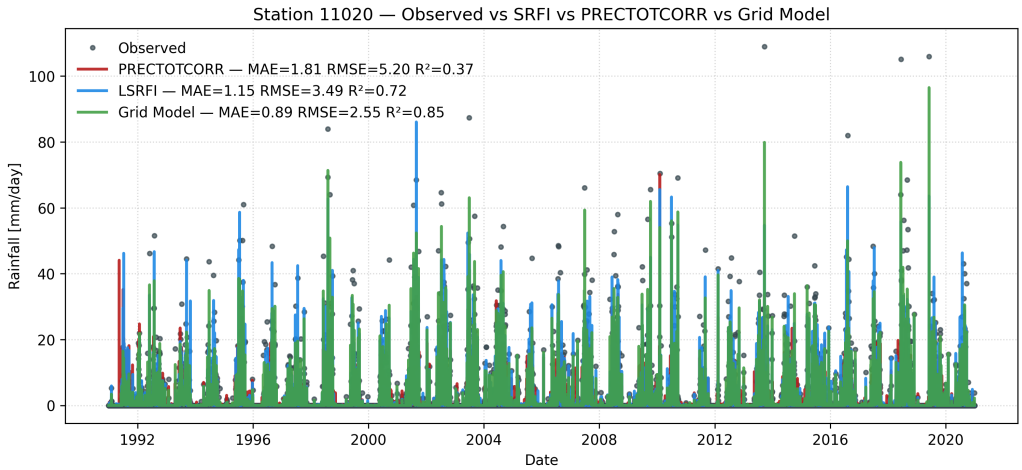

Figure 1: Daily rainfall at station 11020 (México) comparing observations, NASA POWER precipitation (PRECTOTCORR), a local spatial random forest model (SRFI), and a grid-based global model (Grid Model). The plot is generated with MEXsrfdcrPy, and the legend reports MAE, RMSE and R^2 for each series.

Related work

MEXsrfdcrPy sits at the intersection of traditional spatial interpolation, machine-learning-based mapping and open climate-data workflows. It complements geostatistical approaches and global datasets such as WorldClim and CHELSA (Hijmans et al., 2005; Karger et al., 2017) by focusing on the reconstruction of **station series** and by using a single, reusable random forest model trained solely on coordinates and time. The design is inspired by studies demonstrating the strong performance of random forests for spatial and spatio-temporal prediction (Hengl et al., 2018) and by best practices for spatial cross-validation (Roberts et al., 2017). By building on NumPy, pandas and scikit-learn (Harris et al., 2020; McKinney, 2010; Pedregosa et al., 2011) and integrating naturally with Python-based ecosystems for climate and agricultural modeling, MEXsrfdcrPy provides a pragmatic, fully reproducible path from raw, incomplete national station archives to interpolated daily series and high-resolution climate grids.

Acknowledgements

This work was supported by the Secretaría de Ciencia, Humanidades, Tecnología e Innovación (SECIHTI) through a doctoral scholarship to the first author. We acknowledge Colegio de Postgraduados and Universidad Mexiquense del Bicentenario for institutional support. We also thank the International Maize and Wheat Improvement Center (CIMMYT) for fostering collaboration in open climate and agricultural research.

Antonio-Fernández, H., Vaquera-Huerta, H., Rosengaus-Moshinsky, M. M., Pérez-Rodríguez, P., & Crossa, J. (2025a). *Daily meteorological records for operational SMN stations in mexico (1991–2020) with NASA POWER co-located variables (v1.0.0)* [Data set]. Zenodo. https://doi.org/10.5281/zenodo.17636066

Antonio-Fernández, H., Vaquera-Huerta, H., Rosengaus-Moshinsky, M. M., Pérez-Rodríguez, P., & Crossa, J. (2025b). *MissClimatePy: Spatial–temporal imputation for daily climate station records in python (v0.1.1.post1)*. Zenodo. https://doi.org/10.5281/zenodo.17794136

Antonio-Fernández, H., Vaquera-Huerta, H., Rosengaus-Moshinsky, M. M., Pérez-Rodríguez, P., & Crossa, J. (2025c). *SMNdataR: Tools for processing and analyzing mexican meteorological data for climatological applications* (Version v0.1.1). Zenodo. https:

//doi.org/10.5281/zenodo.17495178

Breiman, L. (2001). Random forests. *Machine Learning*, *45*(1), 5–32. https://doi.org/10.1023/A:1010933404324

Comisión Nacional del Agua (CONAGUA). (2012). *Servicio meteorológico nacional: 135 años de historia en méxico*. Secretaría de Medio Ambiente y Recursos Naturales (SEMARNAT).

Harris, C. R., Millman, K. J., Walt, S. J. van der, Gommers, R., Virtanen, P., Cournapeau, D., Wikrent, M., & others. (2020). Array programming with NumPy. *Nature*, *585*, 357–362. https://doi.org/10.1038/s41586-020-2649-2

Hengl, T., Nussbaum, M., Wright, M. N., Heuvelink, G. B. M., & Gräler, B. (2018). Random forest as a generic framework for predictive modeling of spatial and spatio-temporal variables. *PeerJ*, *6*, e5518. https://doi.org/10.7717/peerj.5518

Hijmans, R. J., Cameron, S. E., Parra, J. L., Jones, P. G., & Jarvis, A. (2005). Very high resolution interpolated climate surfaces for global land areas. *International Journal of Climatology*, *25*(15), 1965–1978. https://doi.org/10.1002/joc.1276

Hofstra, N., Haylock, M., New, M., & Jones, P. D. (2008). Comparison of six methods for the interpolation of daily, european climate data. *Journal of Geophysical Research: Atmospheres*, *113*(D21), D21110. https://doi.org/10.1029/2008JD010100

Karger, D. N., Conrad, O., Böhner, J., Kawohl, T., Kreft, H., Soria-Auza, R. W., Zimmermann, N. E., Linder, H. P., & Kessler, M. (2017). Climatologies at high resolution for the earth's land surface areas. *Scientific Data*, *4*, 170122. https://doi.org/10.1038/sdata.2017.122

McKinney, W. (2010). Data structures for statistical computing in python. In S. van der Walt & J. Millman (Eds.), *Proceedings of the 9th python in science conference* (pp. 51–56).

Menne, M. J., Durre, I., Vose, R. S., Gleason, B. E., & Houston, T. G. (2012). An overview of the GHCN-Daily database. *Journal of Atmospheric and Oceanic Technology*, *29*(7), 897–910. https://doi.org/10.1175/JTECH-D-11-00103.1

National Aeronautics and Space Administration. (2020). *NASA POWER project: Global meteorology, surface solar energy and climatology data*. https://power.larc.nasa.gov/

Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., & Duchesnay, É. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, *12*, 2825–2830.

Roberts, D. R., Bahn, V., Ciuti, S., Boyce, M. S., Elith, J., Guillera-Arroita, G., Hauenstein, S., & others. (2017). Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography*, *40*(8), 913–929. https://doi.org/10.1111/ecog.02881

Ruane, A. C., Goldberg, R., & Chryssanthacopoulos, J. (2015). Climate forcing datasets for agricultural modeling: Merged products for gap-filling and historical climate series estimation. *Agricultural and Forest Meteorology*, *200*, 233–248. https://doi.org/10.1016/j.agrformet.2014.09.016

Xu, T., Zhang, Y., Zhang, C., Abodoukayimu, A., & Ma, D. (2024). Deep learning tool: Reconstruction of long missing climate data based on spatio-temporal multilayer perceptron. *Theoretical and Applied Climatology*, *157*, 1235–1250. https://doi.org/10.1007/s00704-024-04945-3

DRAFT